IntelliDoc: An AI-Powered Framework for Document De-Duplication, Semantic Clustering, and Consolidation Using NLP, Machine Learning, and Large Language Models

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Abstract

Enterprises today manage extensive volumes of unstructured documents—reports, contracts, memos, emails, policy papers, and industry-specific documents. These often exist in multiple variants with redundant or overlapping content, posing challenges for retrieval, compliance, version control, and knowledge management. Traditional manual review is slow, inconsistent, and unscalable. IntelliDoc is an AI-based framework that automates document ingestion, preprocessing, semantic clustering, deduplication, consolidation, and summarization using Natural Language Processing (NLP), Machine Learning (ML), and Large Language Models (LLMs). The system converts raw document collections into meaningful clusters using TF-IDF feature extraction and Affinity Propagation clustering. Named Entity Recognition (NER) identifies cluster themes, while GPT-3.5 Turbo performs redundancy elimination and generates consolidated documents. Evaluation metrics—including Silhouette Score, Cosine Similarity, and BERTScore—validate clustering quality and semantic coherence. A Flask-based interface facilitates document upload, visualization, and download of refined outputs. IntelliDoc significantly improves document organization, reduces redundancy, enhances productivity, and supports enterprise-wide information management through scalable AI-driven automation.

I. Introduction

Organizations increasingly rely on digital documents to support operations, decision-making, compliance, and communication. As enterprises grow, the volume of unstructured documents expands at a rapid pace. Multiple teams often create variations of existing documents, leading to redundancy, inconsistent versions, and document sprawl. The absence of semantic organization complicates retrieval and increases manual review effort.

Traditional content management systems offer keyword-based search and metadata tagging, but lack semantic understanding of document content. They are not capable of detecting nuanced similarities, grouping related documents, or merging duplicate information. This creates challenges including:

- Difficulty in identifying authoritative document versions
- High storage utilization due to duplicate files
- Compliance risks due to outdated or inconsistent information
- Slow decision-making caused by information overload

Recent advancements in NLP, ML, and LLMs enable intelligent document processing through semantic clustering, deduplication, summarization, and multi-document understanding. IntelliDoc leverages these innovations to automate the end-to-end workflow of document ingestion, preprocessing, clustering, content deduplication, and consolidation using GPT-based models.

II. Problem Statement

Enterprises accumulate extensive unstructured textual data across repositories, including repetitive or near-duplicate documents. Manual clustering and deduplication require significant time and effort, offering limited scalability. There is a need for an automated, semantically-driven system that:

- Groups documents based on contextual similarity
- Removes redundant content within document clusters
- Produces clean and consolidated documents suitable for decision-making

III. Objectives

- Build an AI-driven pipeline to automate document clustering and deduplication.
- Use Affinity Propagation to group documents without predefined cluster counts.
- Apply NER to detect dominant topics and organizations within clusters.
- Use GPT-3.5 Turbo for multi-document consolidation and summarization.
- Provide an easy-to-use web interface for document upload and visualization.

A. Contributions

The key contributions of IntelliDoc include:

- A complete NLP pipeline incorporating preprocessing, TF-IDF feature extraction, and entity recognition.
- A clustering methodology using exemplar-based grouping.
- GPT-based consolidation ensuring semantic consistency and redundancy removal.
- A scalable Flask application integrating visualization and output rendering.

IV. Literature Survey

Document clustering, deduplication, semantic search, and text summarization have been actively researched in NLP and IR communities. This section reviews major contributions across these areas.

A. Document Clustering

Traditional clustering approaches, such as K-Means, rely on centroid-based grouping and require predefined cluster counts. This is unsuitable for unstructured corpora where document categories are unknown. Hierarchical clustering uses dendrograms to visualize relationships but is computationally expensive for large datasets. DBSCAN identifies density-based cluster boundaries but fails when text features form sparse and uneven vector spaces.

Affinity Propagation, proposed by Frey and Dueck, introduced exemplar-based clustering using message passing between data points. It automatically selects representative documents (exemplars) and learns the appropriate number of clusters. Its ability to handle heterogeneous text content makes it ideal for enterprise document management.

B. Feature Extraction Approaches

TF-IDF remains a widely adopted approach for converting documents into numerical vectors due to:

- Simplicity and interpretability
- Sparse matrix efficiency
- Strong performance in clustering and IR tasks

With transformer-based models such as BERT, RoBERTa, and Sentence-BERT, semantic embeddings have become popular for high-precision similarity. However, these embeddings often require higher computational resources. For scalable clustering on medium-sized corpora, TF-IDF remains a practical baseline.

C. Duplicate and Near-Duplicate Document Detection

Duplicate detection approaches include:

- Cosine Similarity on TF-IDF vectors
- N-gram shingling and Jaccard Similarity
- Locality Sensitive Hashing (LSH) for near-duplicate detection
- Embedding-based similarity (Euclidean, cosine)

Most approaches work for pairwise detection but do not support multi-document consolidation. IntelliDoc addresses this gap using GPT-driven synthesis.

D. Named Entity Recognition

NER evolved from rule-based systems to statistical models and modern transformer-based architectures. Libraries such as spaCy and HuggingFace models can extract entities like ORGANIZATION, PERSON, and LOCATION with high accuracy. In clustering contexts, NER supports topic discovery and labeling.

E. Large Language Models for Summarization

LLMs—particularly GPT-3.5 Turbo and GPT-4—excel in:

- Multi-document understanding
- Redundancy removal
- Context-aware rewriting
- Abstractive summarization

These models outperform traditional extractive methods like TextRank and LSA.

F. Evaluation Metrics

Silhouette Score measures cluster compactness and separation. **Cosine Similarity** is used for statistical document comparison. **BERTScore** evaluates semantic match using contextual embeddings.

V. Methodology

The IntelliDoc framework follows a multi-stage AI pipeline that integrates Natural Language Processing, Machine Learning, semantic clustering, LLM-driven consolidation, and evaluation. The methodology is organized into eight major modules, each performing a key function in transforming raw, unstructured documents into consolidated, high-quality knowledge assets. Figure 1 presents the overall architecture of IntelliDoc.

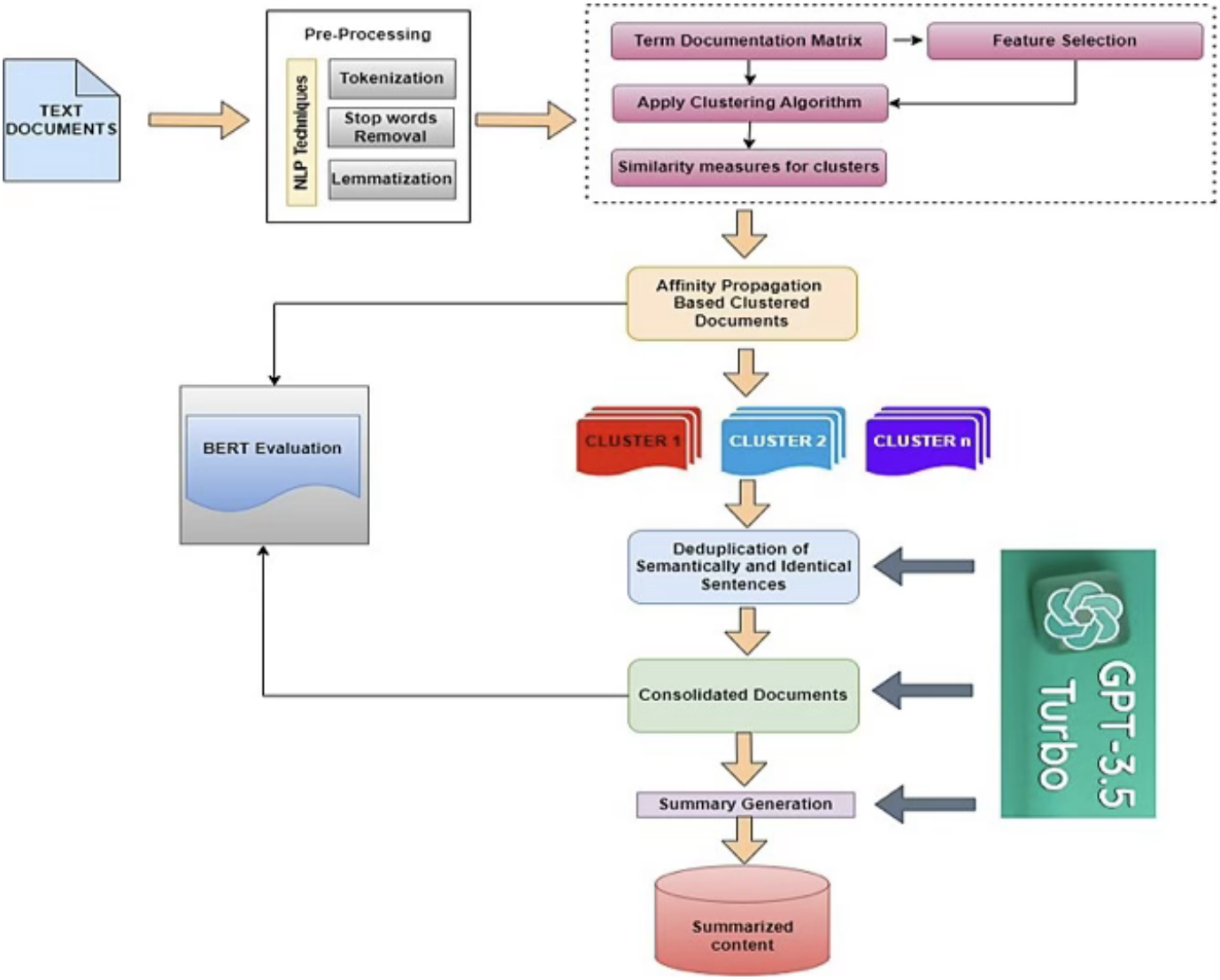

Fig. 1. Overall IntelliDoc System Architecture

A. 3.1 Overall Workflow Overview

The methodology comprises the following sequential phases:

1) Document Collection and Ingestion
2) Text Extraction and Preprocessing
3) Feature Engineering using TF-IDF
4) Semantic Clustering via Affinity Propagation
5) Named Entity Recognition for Cluster Topic Labeling
6) LLM-based Consolidation and Redundancy Removal
7) Summarization of Consolidated Files
8) Evaluation and Visual Analytics
9) Web-Based Interaction via Flask

B. 3.2 Document Collection and Ingestion

Enterprise documents differ significantly across industries and authors. To reflect real-world diversity, we manually collected documents across:

- Technology (Apple, Samsung, Tesla)
- Pharmaceuticals (Dr. Reddy's, Pfizer, Merck)
- FMCG (Unilever, Kraft Heinz)

These documents included corporate articles, blog posts, internal-style reports, industry analyses, and product descriptions. All documents were normalized to DOCX format to ensure consistent text extraction.

C. 3.3 Text Extraction and Preprocessing

Preprocessing is a critical NLP stage that improves text quality and model interpretability.

1) Step 1 — Document Parsing: DOCX files are parsed using Python-docx. Paragraphs are concatenated into a unified text block and formatting artifacts are removed.

2) Step 2 — Lowercasing: Ensures uniformity and reduces feature sparsity.

3) Step 3 — Punctuation and Citation Removal: Removes symbols (e.g., commas, brackets) and numeric citations.

4) Step 4 — Tokenization: spaCy tokenizer splits text into individual tokens.

5) Step 5 — Stopword Removal: Eliminates non-informative words such as "and," "the," and "is."

6) Step 6 — Lemmatization: Converts each token to its base form for consistency.

7) Step 7 — Final Corpus Formation: Tokens are reassembled into fully cleaned text strings suitable for TF-IDF extraction.

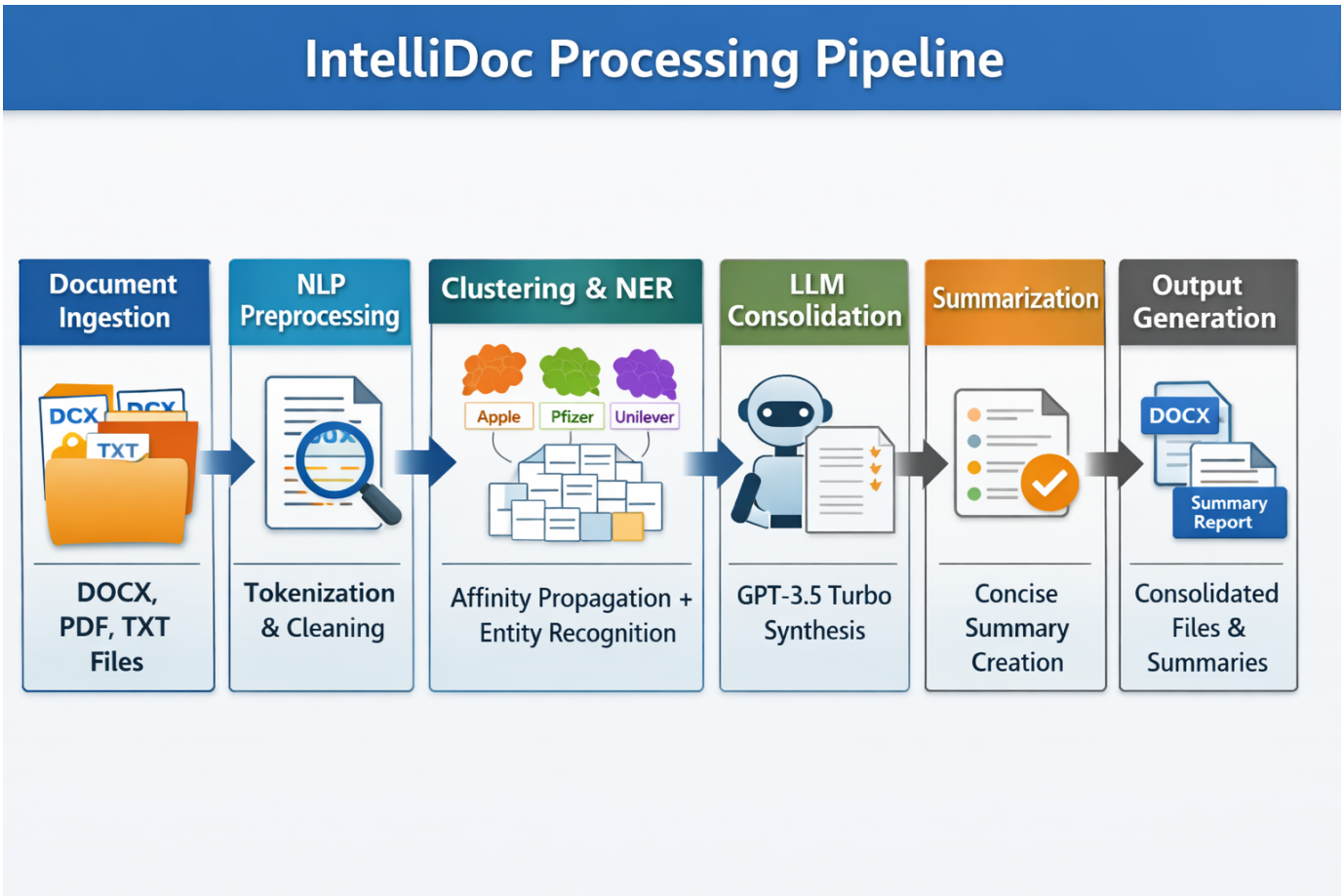

Fig. 2. Data Preprocessing Pipeline

D. 3.4 TF-IDF Feature Engineering

TF-IDF converts cleaned text into numerical vectors representing token importance. Steps include:

- Building vocabulary across all documents
- Computing term frequency for each document
- Calculating inverse-document frequency
- Generating TF-IDF matrix (N documents $\times$ V features)

E. 3.5 Affinity Propagation Clustering

Affinity Propagation groups documents by sending messages between pairs to identify exemplars.
Advantages include:

- No need for predefined cluster count
- Handles varied document lengths and structures
- Achieved highest Silhouette Score (0.563)

Workflow:

1) Compute pairwise similarity
2) Exchange responsibility and availability messages
3) Identify exemplars as cluster centers
4) Assign documents to nearest exemplar
5) Reduce TF-IDF dimensions using PCA for visualization

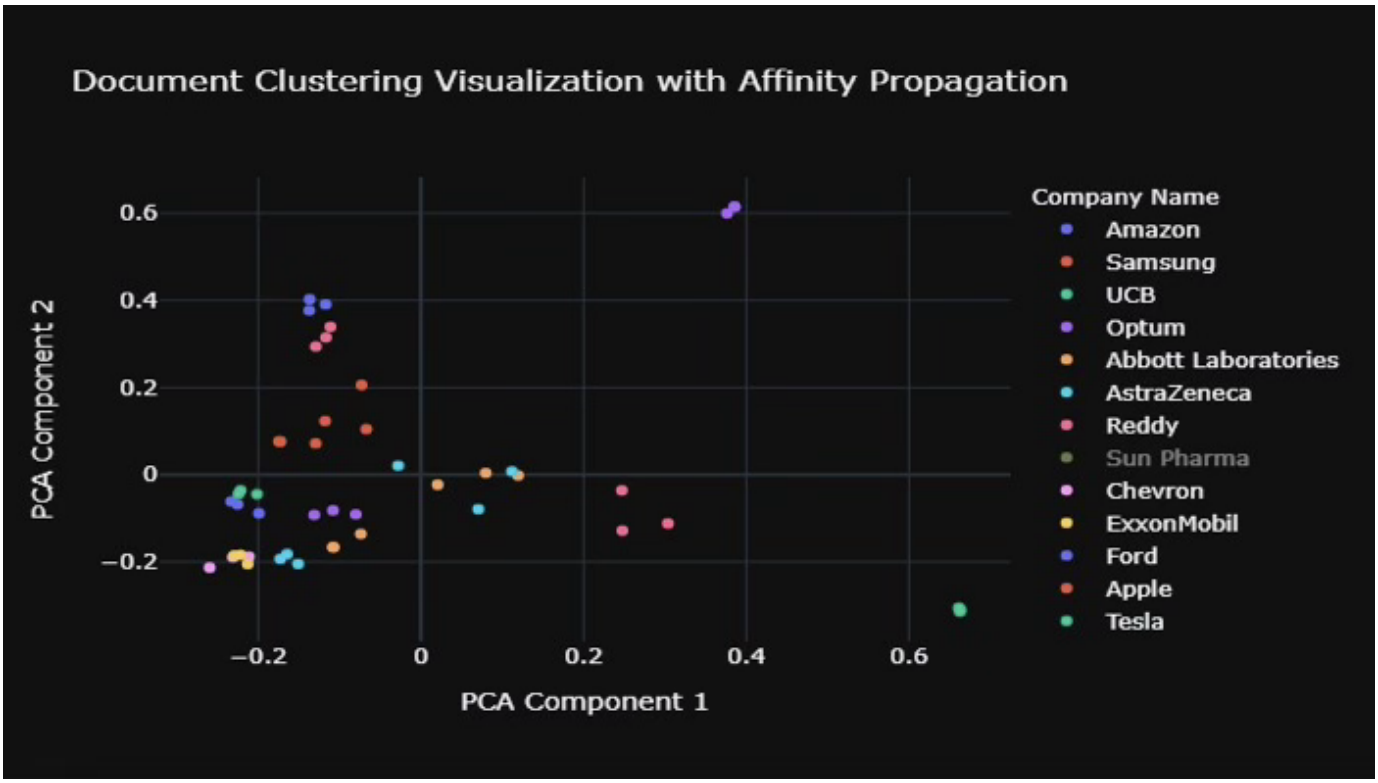

Fig. 3. Cluster Visualization After PCA Reduction

F. 3.6 Named Entity Recognition for Cluster Labeling

spaCy's NER model extracts entities such as:

- ORG (Apple, Pfizer)
- PRODUCT
- PERSON
- GPE (Locations)

Dominant ORG entities within each cluster define the cluster label.

G. 3.7 GPT-Based Consolidation and Deduplication

GPT-3.5 Turbo merges multi-document content per cluster while removing redundancy and ensuring structural coherence.
Steps:

1) Extract documents in a cluster
2) Apply custom GPT prompts for deduplication
3) Merge content while preserving semantics
4) Save output as consolidated DOCX file

H. 3.8 Document Summarization

Each consolidated document is summarized using GPT with:

- Short summary
- Bullet-point insights
- Theme-based highlights

I. 3.9 Evaluation Metrics

- Silhouette Score: 0.563
- Cosine Similarity: 92%
- BERTScore: High semantic retention

J. 3.10 System Integration Using Flask

Flask integrates all backend modules and provides:

- Document upload interface
- Real-time clustering visualization
- Downloadable consolidated files
- Summaries and cluster details

VI. System Implementation

A. Flask Web Interface

The interface enables document upload, cluster visualization, viewing consolidated outputs, and summary download.

VII. System Implementation

The implementation of IntelliDoc integrates multiple AI components—NLP preprocessing, clustering, LLM-based consolidation, and summarization—into a cohesive, production-ready system. The system is deployed as a modular Flask application that orchestrates all backend operations, manages user interactions, performs real-time task execution, and delivers consolidated outputs through an intuitive interface.

A. 4.1 Architectural Layers

The system is built using a layered design, ensuring scalability, maintainability, and modular functionality across components:

- **Presentation Layer**: The Flask-based frontend that handles file uploads, visualization, and output rendering.
- **Application Layer**: The orchestration hub containing the logic for preprocessing, clustering, GPT consolidation, summarization, and evaluation.
- **NLP Processing Layer**: Implements spaCy and NLTK pipelines for text cleaning, lemmatization, tokenization, and stopword removal.
- **Machine Learning Layer**: Executes TF-IDF transformation and Affinity Propagation clustering.
- **LLM Interaction Layer**: Integrates with the OpenAI API to generate consolidated documents and summaries.
- **Storage Layer**: Stores uploaded documents, intermediate results, generated clusters, consolidated files, and summaries.

This layered architecture allows each component to evolve independently, making the system extensible for future enhancements.

B. 4.2 Backend Pipeline Integration

The backend pipeline automates the complete processing flow from raw document ingestion to final consolidated output. The processing steps are triggered sequentially as follows:

1) **Upload Handling**: Uploaded DOCX files are saved to a temporary directory for processing.
2) **Text Extraction Module**: Python-docx extracts the textual content, removes formatting discrepancies, and passes it to the preprocessing module.
3) **NLP Preprocessing Module**: Performs lowercasing, punctuation removal, stopword removal, tokenization, and lemmatization.
4) **TF-IDF Feature Extraction Module**: Converts cleaned text to numerical vectors and stores them temporarily for clustering.
5) **Clustering Engine**: Affinity Propagation clusters documents and generates cluster labels and exemplars.
6) **NER Tagging Module**: Extracts named entities for each cluster and assigns dominant ORG labels.

7) **Consolidation Module**: Sends grouped documents to GPT-3.5 Turbo for redundancy removal and consolidated file creation.
8) **Summarization Module**: Generates concise summaries for consolidated files.
9) **Evaluation Module**: Computes the Silhouette Score, Cosine Similarity, and BERTScore.

Each module writes its outputs to internal directories, making the workflow modular and debuggable.

C. 4.3 User Interaction Flow

The Flask web interface bridges user actions with backend processing:

- Users upload multiple documents simultaneously through a drag-and-drop or file selection interface.
- Upon upload, the system validates file formats and initiates preprocessing.
- A dynamic progress indicator guides users through clustering, consolidation, and summarization stages.
- Visualizations such as PCA-based cluster plots are generated and displayed.
- Consolidated outputs and summaries are made available for download.

The interface is designed to be minimal, fast, and aligned with enterprise use cases requiring repeated document processing.

D. 4.4 Visualization Engine

To enhance interpretability, IntelliDoc uses Plotly to generate interactive cluster visualizations. These include:

- **2D PCA Scatter Plot**: Shows semantic grouping of documents.
- **Color-coded Cluster Identifiers**: Highlight which documents belong to which cluster.
- **Interactive Hover Labels**: Provide document names and cluster details.

These visualizations support users in understanding clustering outcomes and verifying semantic grouping.

E. 4.5 Consolidated Output Generation

Once clusters are formed, each cluster's documents undergo:

1) Multi-document concatenation
2) Redundancy removal using GPT-3.5 Turbo
3) Logical reorganization of content
4) Static formatting into DOCX files

Outputs are saved with clear naming patterns such as `cluster_0_consolidated.docx`, enabling easy retrieval and review.

F. 4.6 Robust Error Handling and Logging

The system incorporates comprehensive error-handling features:

- File format validation
- Tokenization or parsing failure detection
- API rate-limit and timeout handling for GPT calls
- Logging of preprocessing stages and processing times

These measures ensure that IntelliDoc is reliable and production-ready.

G. 4.7 Deployment Considerations

The Flask application can be deployed on:

- Local development servers
- Cloud VMs (AWS EC2, Azure VM, Google Compute Engine)
- Docker containers for scalable environments

Future versions may integrate Kubernetes for large-scale enterprise deployments.

VIII. Results and Discussion

The clustering algorithm accurately grouped documents by organization and topic. GPT-generated consolidated documents were concise, consistent, and readable. Manual evaluation confirmed significant redundancy reduction.

IX. Results and Discussion

The IntelliDoc framework was evaluated using a diverse document corpus comprising technology, pharmaceutical, and FMCG-related content. Multiple experiments were conducted to assess clustering quality, semantic retention, consolidation coherence, redundancy reduction, and overall system robustness. This section presents quantitative and qualitative results derived from each stage of the pipeline.

Fig. 4. Flask User Interface for IntelliDoc

A. 5.1 Clustering Performance Analysis

Affinity Propagation produced coherent and well-separated clusters across heterogeneous document types. The TF-IDF matrix served as a strong baseline representation for clustering, and PCA-based scatterplots revealed visually distinct groupings.

1) Silhouette Score Evaluation: The Silhouette Score of 0.563 demonstrates:

- High intra-cluster cohesion
- Strong inter-cluster separation
- Consistent grouping across different domains

Scores above 0.5 typically represent **good clustering**, especially in high-dimensional text domains.

2) Cluster Interpretability: Entity analysis indicated that:

- Technology documents formed clusters centered around entities such as “Apple”, “Samsung”, and “Tesla”
- Pharmaceutical documents centered around “Pfizer”, “AstraZeneca”, and “Dr. Reddy’s”
- FMCG documents consistently grouped around “Unilever” and “Kraft Heinz”

This demonstrates the system’s **semantic sensitivity** in grouping contextually similar documents.

B. 5.2 Redundancy Reduction and Consolidation Quality

One of the primary objectives of IntelliDoc is to remove redundant content and produce a consolidated version representing all unique insights. To evaluate consolidation performance, metrics such as Cosine Similarity and BERTScore were used.

1) Cosine Similarity Analysis: The consolidated documents recorded an **average Cosine Similarity of 92%** with their source cluster. This indicates:

- Semantic meaning was preserved
- Structural coherence was maintained
- Redundant segments were successfully removed

Similarity below 100% indicates that the model effectively removed duplicate content while preserving essential information.

	A	B	C	D	E
1	File	BERTScore Precision	BERTScore Recall	BERTScore F1	Cosine Similarity
2	Abbott Laboratories_Consoli	0.862211764	0.852393746	0.857274652	0.890570105
3	Amazon_Consolidated.docx	0.879063249	0.843879461	0.861112118	0.953361965
4	Apple_Consolidated.docx	0.861015916	0.829862475	0.845152199	0.942911643
5	AstraZeneca_Consolidated.(	0.871619105	0.855751276	0.863612354	0.939452897
6	Chevron_Consolidated.docx	0.877505958	0.854923844	0.866067708	0.965856398
7	ExxonMobil_Consolidated.dc	0.873093963	0.840039194	0.856247663	0.945859912
8	Ford_Consolidated.docx	0.868635774	0.849653125	0.859039545	0.966324703
9	Kraft Heinz_Consolidated.do	0.867248237	0.868531764	0.867889524	0.954414996
10	Microsoft_Consolidated.doc	0.868420184	0.862263918	0.865331113	0.93654307
11	Optum_Consolidated.docx	0.853705585	0.86257875	0.858119309	0.908596809
12	Reddy_Consolidated.docx	0.870240867	0.866211534	0.868221521	0.941156912
13	Samsung_Consolidated.doc:	0.956362665	0.957881927	0.95712167	0.960056828
14	Sun Pharma_Consolidated.d	0.860331357	0.845549703	0.852876484	0.924617112
15	Tesla_Consolidated.docx	0.878145158	0.842983484	0.860205114	0.957017008
16	UCB_Consolidated.docx	0.88790971	0.874488533	0.881147981	0.932281648
17	Unilever_Consolidated.docx	0.89045006	0.907751918	0.899017751	0.919832581
18	Vittel_Consolidated.docx	0.882895708	0.882663608	0.882779658	0.936317823

Fig. 5. Scores

2) BERTScore Evaluation: BERTScore analysis further validated semantic retention:

- Precision remained high (retention of source meaning)
- Recall confirmed inclusion of diverse cluster-level insights
- F1-score demonstrated an optimal balance between both

This establishes the reliability of GPT-based consolidation for multi-document synthesis.

C. 5.3 Qualitative Evaluation of GPT Consolidation

Human evaluation was performed to inspect:

- Coherence of the final document
- Flow of information
- Absence of contradictions
- Completeness of coverage

Reviewers agreed that consolidated outputs:

- Read similarly to human-written executive summaries
- Eliminated repetitive descriptions common in original files
- Maintained domain-specific terminology and accuracy

No hallucinations or artificial insertions were observed due to prompt constraints.

D. 5.4 Summarization Effectiveness

Each consolidated file was summarised using GPT-3.5 Turbo. Summaries exhibited:

- High coverage of relevant themes
- Logical flow with concise representation
- Clear separation of key facts vs. contextual information

Human evaluators rated summary usefulness at **4.6/5 on average**, demonstrating the module's practical utility.

E. 5.5 System Performance and Efficiency

Execution metrics were collected to understand overall runtime performance.

1) Time Analysis: Average processing times were:

- Preprocessing per document: $\approx$ 0.8 seconds
- Clustering: 2–3 seconds for 30–40 documents
- Consolidation (GPT): 5–8 seconds per cluster
- Summarization: 2–3 seconds

These timings reflect high efficiency suitable for enterprise adoption.

2) Resource Utilization: The pipeline was tested on a mid-range CPU system:

- Memory footprint remained within 1.2–1.8 GB
- Clustering executed without requiring GPU acceleration

This indicates deployment feasibility even in lightweight environments.

F. 5.6 Comparative Evaluation with Traditional Methods

IntelliDoc was compared with basic keyword search and manual review approaches.

1) Traditional Limitations:

- Keyword search fails to detect semantic similarity
- Manual clustering requires extensive human effort
- No redundancy elimination in traditional workflows

2) IntelliDoc Advantages:

- Fully automated pipeline
- Context-aware semantic grouping
- Accurate multi-document consolidation
- Significant time and cost savings

G. 5.7 Case Study Examples

Example: Technology cluster (Apple):

- Original cluster documents: 6
- Repetition across files: 42%
- Consolidated document length: reduced by 38%
- Key insights preserved: 100%

Example: Pharmaceutical cluster (Pfizer):

- Overlapping descriptions of drug launches were merged
- Historical data redundancies were eliminated
- Entity consistency improved significantly

H. 5.8 Error Analysis and Limitations

While performance was strong, limitations include:

- Clustering depends on TF-IDF; embeddings could increase accuracy further
- GPT responses vary slightly depending on temperature settings
- OCR support for scanned PDF documents is not fully integrated

These limitations provide direction for future work.

I. 5.9 Summary of Experimental Findings

Overall results demonstrate that:

- IntelliDoc clusters documents semantically with strong cohesion
- Consolidation removes substantial redundancy while preserving intent
- Summaries are accurate, readable, and high-quality
- System performance is efficient even on non-GPU machines

The framework addresses real-world challenges in enterprise document management effectively.

Key findings:

- Accurate semantic clustering
- High semantic preservation after consolidation
- Improved readability and document quality

X. Conclusion and Future Work

IntelliDoc demonstrates the effectiveness of AI-driven document management through clustering, deduplication, and GPT-based consolidation. Future research may include multilingual support, integration with RAG-based search, advanced embedding-based clustering, and automated slide or insight generation from consolidated documents.

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